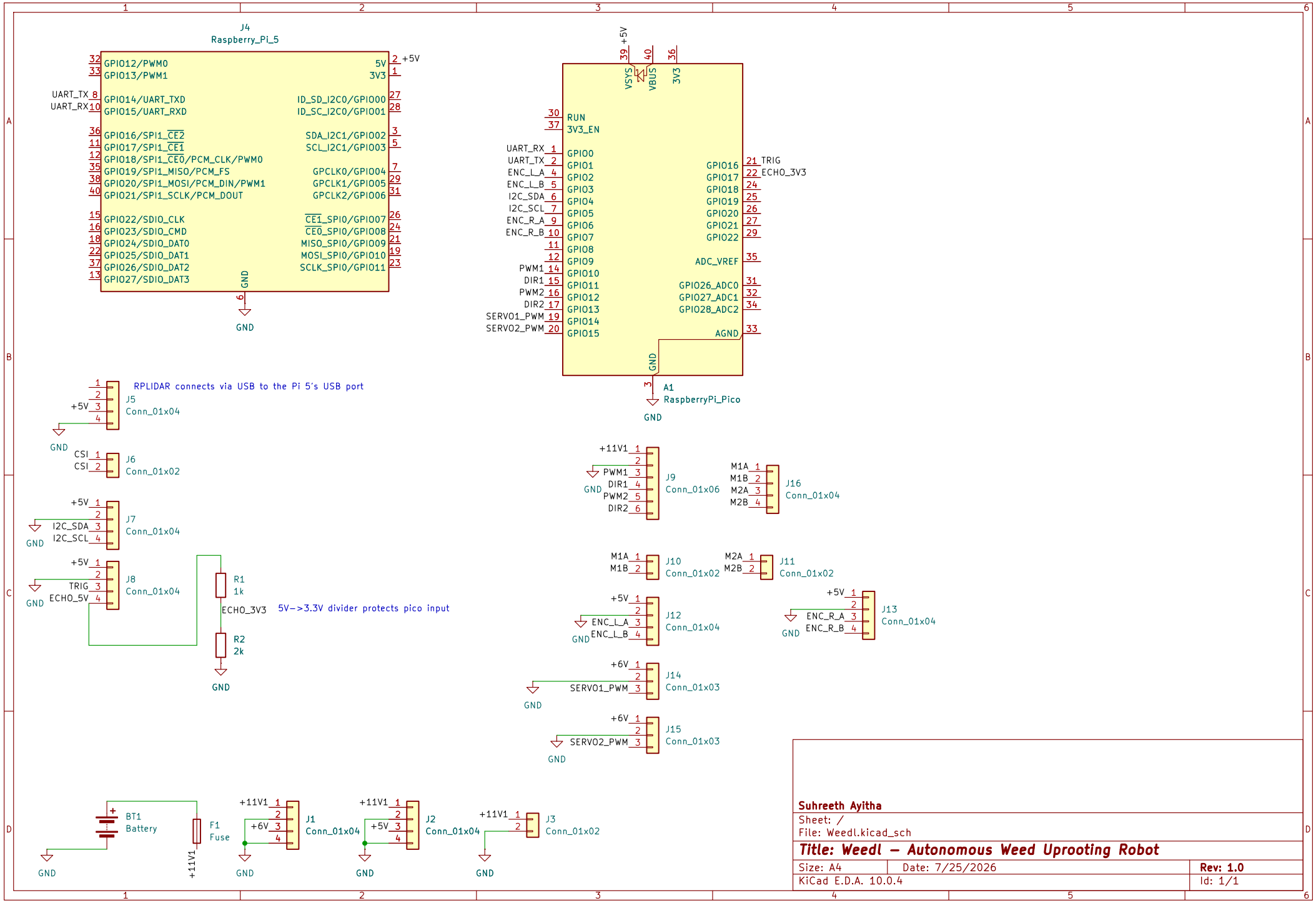

J4
Raspberry_Pi_5
GPIO12/PWM0
GPIO13/PWM1
GPIO14/UART_TXD
GPIO15/UART_RXD
GPIO16/SPI1_CE2
GPIO17/SPI1_CE1
GPIO18/SPI1_CE0/PCM_CLK/PWM0
GPIO19/SPI1_MISO/PCM_FS
GPIO20/SPI1_MOSI/PCM_DIN/PWM1
GPIO21/SPI1_SCLK/PCM_DOUT
GPIO22/SDIO_CLK
GPIO23/SDIO_CMD
GPIO24/SDIO_DAT0
GPIO25/SDIO_DAT1
GPIO26/SDIO_DAT2
GPIO27/SDIO_DAT3
5V
3V3
ID_SD_I2C0/GPIO00
ID_SC_I2C0/GPIO01
SDA_I2C1/GPIO02
SCL_I2C1/GPIO03
GPCLK0/GPIO04
GPCLK1/GPIO05
GPCLK2/GPIO06
CE1_SPI0/GPIO07
CE0_SPI0/GPIO08
MISO_SPI0/GPIO09
MOSI_SPI0/GPIO10
SCLK_SPI0/GPIO11
UART_TX
UART_RX
GND
+5V
VSYS
VBUS
3V3
RUN
3V3_EN
GPIO0
GPIO1
GPIO2
GPIO3
GPIO4
GPIO5
GPIO6
GPIO7
GPIO8
GPIO9
GPIO10
GPIO11
GPIO12
GPIO13
GPIO14
GPIO15
GPIO16
GPIO17
GPIO18
GPIO19
GPIO20
GPIO21
GPIO22
ADC_VREF
GPIO26_ADC0
GPIO27_ADC1
GPIO28_ADC2
AGND
TRIG
ECHO_3V3
A1
RaspberryPi_Pico
RPLIDAR connects via USB to the Pi 5's USB port
J5
Conn_01x04
J6
Conn_01x02
CSI
J7
I2C_SDA
I2C_SCL
J8
TRIG
ECHO_5V
R1
1k
R2
2k
5V->3.3V divider protects pico input
+11V1
PWM1
DIR1
PWM2
DIR2
J9
Conn_01x06
M1A
M1B
M2A
M2B
J16
J10
J11
J12
ENC_L_A
ENC_L_B
J13
ENC_R_A
ENC_R_B
+6V
SERVO1_PWM
J14
Conn_01x03
SERVO2_PWM
J15
BT1
Battery
F1
Fuse
J1
J2
J3
Suhreeth Ayitha
Sheet: /
File: Weedl.kicad_sch
Title: Weedl – Autonomous Weed Uprooting Robot
Size: A4
Date: 7/25/2026
Rev: 1.0
KiCad E.D.A. 10.0.4
Id: 1/1